

NAME: _____

CLASS: _____



DANEBANK
An Anglican School for Girls

2008
Year 12
Half Yearly Examination

Mathematics

Extension 1

Outcomes Examined: H3, H5, H8, PE2, PE3, PE5, HE2, HE5, HE6, HE7

Weighting of Task: 30%

General Instructions

- Reading time – 5 minutes
- Working time – $1\frac{1}{2}$ hours
- Write using black or blue pen
- All necessary working should be shown in every question otherwise full marks may not be awarded.
- Board-approved calculators may be used
- Start each new question in a new booklet

Total Marks – 63

Attempt All Questions 1 - 5

Questions are NOT of equal value

This paper MUST NOT be removed from the examination room

Total marks – 63
Attempt Questions 1- 5
All questions are not of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (13 marks) Use a SEPARATE writing booklet.

Marks

- (a) If $x = 2 + \sqrt{3}$, find the exact value of $x + \frac{1}{x}$. Hence, or otherwise find the exact value of $x^2 + \frac{1}{x^2}$. 3
- (b) Find the ratio by which the point $C(3,10)$ divides the line segment from $A(2,8)$ to $B(6,16)$. 2
- (c) Solve $\left| \frac{x}{x+1} \right| \geq \frac{1}{2}$ 3
- (e) Find the acute angle between the line $y = 3x + 1$ and $2y = 1 + x$. 2
- (f) If $\frac{d^2y}{dx^2} = 2e^x - 3e^{-x}$ and when $x = 0$, $y = 6$ and $\frac{dy}{dx} = 5$, find an expression for y in terms of x . 3

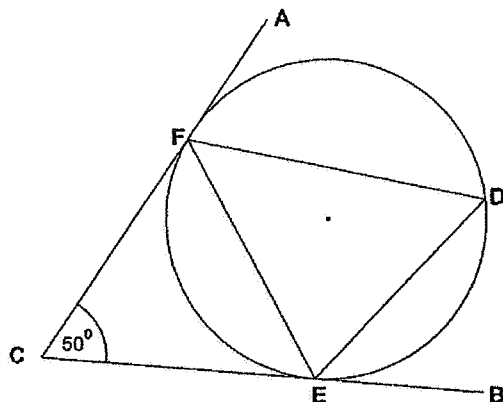
Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Solve the equation $\sin 2x = \tan x$ for $0 \leq x \leq \pi$.

3

- (b)



Copy this diagram in your answer booklet

In the diagram, AC and BC are tangents to the circle, touching the circle at F and E respectively. $\angle ACB$ equals 50° .

- (i) Show the $\angle CEF = 65^\circ$. **2**
- (ii) Hence find $\angle EDF$. **1**
- (c) Two of the roots of the equation $x^3 + mx^2 + nx + l = 0$ are equal in value but opposite in sign.
- (i) Prove that $x = -m$ is a solution for the above equation. **2**
- (ii) Hence, show that $l = mn$. **1**
- (d) Consider the curve $y = 2 \sin 2x$
- (i) Graph this curve for the domain $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ **2**
- (ii) Explain, by the use of this graph, why the area under $y = 2 \sin 2x$ from $x = -\frac{\pi}{2}$ to $x = \frac{\pi}{2}$ can be found by $A = 2 \int_0^{\frac{\pi}{2}} 2 \sin 2x dx$ **1**

Question 3 (13 marks) Use a SEPARATE writing booklet.

Marks

- (a) Prove by Mathematical Induction that $5^{2n} + 12^{n-1}$ is divisible by 13 for all positive integers n . 3

- (b) Given $f(x) = \sin^4 3x$

- (i) Find $f'(x)$ 1

- (ii) Show that $12 \cos 3x \sin^3 3x = 6 \sin 6x \sin^2 3x$ 1

- (ii) Hence $\int \sin 6x \sin^2 3x dx$ 1

- (c) It is given that $P(x) = 2x^4 - 5x^3 + 5x^2 - 4x - 4$.

- (i) Show that 2 and $-\frac{1}{2}$ are zeros of $P(x)$. 1

- (ii) Express $P(x)$ as the product of three algebraic factors and hence show that $P(x) = 0$ has only two real roots. 2

- (iii) Hence, find the set of values of x for $P(x) > 0$. 1

- (d) John was asked to find $\int_0^1 \frac{x}{\sqrt{1+x^2}} dx$, using the substitution $u = 1 + x^2$. 3

Below is his solution. Clearly identify the mistakes and explain why they are wrong.

$$u = 1 + x^2 \quad \therefore \quad \frac{du}{dx} = 2x$$

$$\frac{du}{2} = x dx$$

$$\therefore \int_0^1 \frac{x}{\sqrt{1+x^2}} dx = \int_0^1 \frac{1}{2\sqrt{u}} du$$

$$= \int 2u^{\frac{1}{2}} du$$

$$= \left[u^{\frac{-1}{2}} \right]_0^1$$

$$= (1) - (0)$$

$$= 1 \text{ units}^2$$

Question 4 (13 marks) Use a SEPARATE writing booklet.

Marks

- (a) Prove that $\frac{1 - \cos x}{\sin x} = t$, where $t = \tan \frac{x}{2}$. 2
- (b) Find $\int \sin x \cos^3 x \, dx$, using the substitution $u = \cos x$. 2
- (c) (i) Given $y = \sin x + \cos x$, show that $y^2 = 1 + \sin 2x$ 2
- (ii) Find the exact volume of the solid which is formed by rotating $y = \sin x + \cos x$ about the x -axis, from $x = 0$ to $x = \frac{\pi}{2}$. 2
- (d) (i) Sketch the curves $y = \cos x$ and $y = \cos^2 x$ on the same diagram, for $0 \leq x \leq \frac{\pi}{2}$. 2
- (ii) Find the exact area between these two curves. 3

Ext. 1. Solⁿ. 2008 H-4 rly

1. a) $x = 2 + \sqrt{3}$

i) $x + \frac{1}{x} = 2 + \sqrt{3} + \frac{1}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}}$

$= 2 + \sqrt{3} + \frac{2 - \sqrt{3}}{1}$
 $= 4$

ii) $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ $\left\{ \left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 \right\}$

$= 4^2 - 2$
 $= 14$

b) $AC = \sqrt{(3-2)^2 + (10-8)^2}$
 $= \sqrt{5}$

$CB = \sqrt{(6-3)^2 + (16-10)^2}$
 $= \sqrt{45} = 3\sqrt{5}$

$AC : CB = \sqrt{5} : 3\sqrt{5}$
 $= 1 : 3$

c) $\left| \frac{x}{x^2+1} \right| \geq \frac{1}{2}$

square both sides

$\frac{x^2}{(x^2+1)^2} \geq \frac{1}{4}$

$4x^2 \geq (x^2+1)^2$

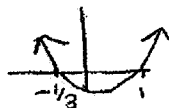
$4x^2 - (x^2 + 2x + 1) \geq 0$

$3x^2 - 2x - 1 \geq 0$

$(3x+1)(x-1) \geq 0$

$x \leq -\frac{1}{3}, x \geq 1$

but $x \neq -1$



e) $y = 3x + 1$ $m_1 = 3$

$2y = 1 + x$ $m_2 = \frac{1}{2}$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{3 - 1/2}{1 + 3 \times 1/2} \right|$

$= 1$

$\theta = 45^\circ$

f) $\frac{d^2 y}{dx^2} = 2e^x - 3e^{-x}$

$\frac{dy}{dx} = \int (2e^x - 3e^{-x}) dx$

$= 2e^x + 3e^{-x} + C$ but $\frac{dy}{dx} = 5$ when $x = 0$

$\therefore 5 = 2e^0 + 3e^0 + C$

$C = 0$

$y = \int (2e^x + 3e^{-x}) dx$

$= 2e^x - 3e^{-x} + C$ but $y = 6$ when $x = 0$

$\therefore 6 = 2e^0 - 3e^0 + C$

$C = 7$

$\therefore y = 2e^x - 3e^{-x} + 7$

Q2.

a) $\sin 2x = \tan x$

$2 \sin x \cos x = \frac{\sin x}{\cos x}$

$2 \sin x \cos^2 x - \sin x = 0$

$\sin x (2 \cos^2 x - 1) = 0$

$\therefore \sin x = 0$ $2 \cos^2 x - 1 = 0$

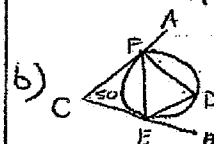
$x = 0, \pi$

$\cos x = \pm \frac{1}{\sqrt{2}}$

$x = \frac{\pi}{4}, \pi - \frac{\pi}{4}$

$\therefore x = 0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi$

4th



i) $CE = CF$ (tangents from ext. pt are =)

$\therefore \triangle CEF$ is isosceles

$2 \angle CEF + 50 = 180$ ($\angle$ sum of Δ)

$\angle CEF = 130 \div 2$

$= 65^\circ$

ii) $\angle CEF = \angle EDF = 65^\circ$ (angle in alternate)

2c.) $x^3 + mx^2 + nx + l = 0$

roots are $\alpha, -\alpha$ & β

$\therefore$ Sum of roots $= -\frac{b}{a}$

$\alpha + -\alpha + \beta = -m$

$\beta = -m$

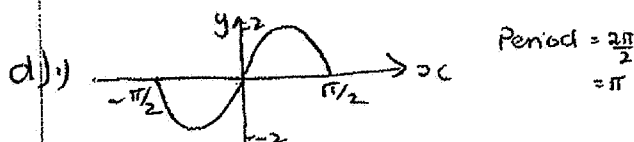
$\therefore x = -m$ is a solⁿ to the eqⁿ.

ii) When $x = -m$

$(-m)^3 + m(-m)^2 + n(-m) + l = 0$

$-m^3 + m^3 - mn + l = 0$

$l = mn$



ii) Since $y = 2 \sin 2x$ is odd $\therefore$ area from $-\frac{\pi}{2}$ to 0 is = but opp in sign to area from $\frac{\pi}{2}$ to 0

$\therefore A = 2 \int_0^{\pi/2} 2 \sin 2x dx$

Q3. a) Test $n=1$

$5^2 + 12^0 = 25 + 1$

$= 26$ which is divisible by 13.

$\therefore$ true for $n=1$

Assume true for $n=k$

$5^{2k} + 12^{k-1} = 13Q$ where Q is an integer

$\therefore 5^{2k} = 13Q - 12^{k-1}$

Prove true for $n=k+1$

Now $5^{2(k+1)} + 12^{k+1-1} = 5^{2k+2} + 12^k$
 $= 5^2 \cdot 5^{2k} + 12^k$

$= 25(13Q - 12^{k-1}) + 12^k$

$= 25 \cdot 13Q - 25 \cdot 12^{k-1} + 12^k$

$= 25 \cdot 13Q - 25 \cdot 12^{k-1} + 12 \cdot 12^{k-1}$

$= 25 \cdot 13Q - 13 \cdot 12^{k-1}$
 $= 13(25Q - 12^{k-1})$

which is divisible by 13 since k is a positive integer.

etc

b) $f(x) = \sin^4 3x$
 $= (\sin 3x)^4$

i) $f'(x) = 4 \sin^3 3x \cdot 3 \cos 3x$
 $= 12 \cos 3x \sin^3 3x$

ii) $12 \cos 3x \sin^3 3x$
 $= 12 \cos 3x \cdot \sin 3x \cdot \sin^2 3x$
 $= 6 \times 2 \cos 3x \sin 3x \cdot \sin^2 3x$
 $= 6 \sin 6x \cdot \sin^2 3x$

iii) $\therefore \int \sin 6x \sin^2 3x dx = \int 2 \cos 3x \sin^3 3x dx$
 $= \frac{1}{6} \sin^4 3x + C$

c) $P(x) = 2x^4 - 5x^3 + 5x^2 - 4x - 4$

i) $P(2) = 2 \cdot 2^4 - 5 \cdot 2^3 + 5 \cdot 2^2 - 4 \cdot 2 - 4$
 $= 0$

$P(-\frac{1}{2}) = 2(-\frac{1}{2})^4 - 5(-\frac{1}{2})^3 + 5(-\frac{1}{2})^2 - 4(-\frac{1}{2}) - 4$
 $= 0$

$\therefore$ Since $P(x)$ is 0 in both cases $\therefore$ they are zeros.

ii) $(x-2)(2x+1)$ are factors

$\therefore (2x^2 - 3x - 2)$ is a factor

$(2x^2 - 3x - 2) \overline{) 2x^4 - 5x^3 + 5x^2 - 4x - 4}$
 $\underline{2x^4 - 3x^3 - 2x^2}$
 $-2x^3 + 7x^2 - 4x$
 $\underline{-2x^3 + 3x^2 + 2x}$
 $4x^2 - 6x - 4$
 $\underline{4x^2 - 6x - 4}$
 0

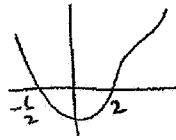
but $x^2 - x + 2$

$\Delta = (-1)^2 - 4 \cdot 1 \cdot 2$
 < 0

$\therefore$ true def so no roots.

$$iii) (2x+1)(x-2)(x^2-x+2) > 0$$

↑
+ve def



$$\therefore (2x+1)(x-2) > 0$$

$$x < -\frac{1}{2} \text{ or } x > 2$$

$$d) \int_0^1 \frac{1}{2\sqrt{u}} du \quad - \text{limits have not changed}$$

$$\int 2u^{1/2} du \quad - \frac{1}{2\sqrt{u}} \text{ is not } 2u^{1/2}$$

- no limits of int.

$$= \left[u^{-1/2} \right]_0^1 \quad \text{limits still wrong this has been diff. not integrated.}$$

$$= 1 - 0 \quad - 0^{-1/2} \text{ is undefined not } 0.$$

units² - not an area

$$4a) \frac{1 - \cos x}{\sin x} = t$$

$$\therefore \text{LHS} = 1 - \frac{(1-t^2)}{(1+t^2)}$$

$$= \frac{\frac{2t}{1+t^2}}{\frac{1+t^2}{1+t^2}} \times \frac{1+t^2}{2t}$$

$$= \frac{2t^2}{2t}$$

$$= t$$

= RHS

$$b) u = \cos x$$

$$\frac{du}{dx} = -\sin x$$

$$-du = \sin x dx$$

$$\int \sin x \cos^3 x dx = \int -u^3 du$$

$$= -\frac{u^4}{4} + C$$

$$= -\frac{\cos^4 x}{4} + C$$

$$c) i) y = \sin x + \cos x$$

$$y^2 = (\sin x + \cos x)^2$$

$$= \sin^2 x + \cos^2 x + 2 \sin x \cos x$$

$$= 1 + \sin 2x$$

ii)

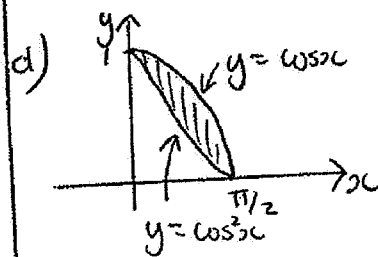
$$\text{Vol} = \pi \int_0^{\pi/2} y^2 dx$$

$$= \pi \int_0^{\pi/2} (1 + \sin 2x) dx$$

$$= \pi \left[x - \frac{\cos 2x}{2} \right]_0^{\pi/2}$$

$$= \pi \left\{ \left(\frac{\pi}{2} - \frac{\cos \pi}{2} \right) - \left(0 - \frac{\cos 0}{2} \right) \right\}$$

$$= \pi \left(\frac{\pi}{2} + 1 \right) \text{ units}^3$$



$$ii) \text{Area} = \int_0^{\pi/2} (\cos x - \cos^2 x) dx$$

$$= \int_0^{\pi/2} \left(\cos x - \frac{1}{2} [\cos 2x + 1] \right) dx$$

$$= \left[\sin x - \frac{1}{4} \sin 2x - \frac{1}{2} x \right]_0^{\pi/2}$$

$$= \left(\sin \frac{\pi}{2} - \frac{1}{4} \sin \pi - \frac{1}{2} \cdot \frac{\pi}{2} \right) - \left(\sin 0 - \frac{1}{4} \sin 0 - \frac{1}{2} \cdot 0 \right)$$

$$= 1 - \frac{\pi}{4} \text{ units}^2$$

Q5 a) Period = $\frac{2\pi}{n}$ amp = 4
 $= 24$

$$\frac{2\pi}{n} = 24$$

$$n = \frac{2\pi}{24}$$

$$= \pi/12$$

$$\therefore e_1^7 y = 4 \cos\left(\frac{\pi x}{12}\right)$$

$$ii) y = 4 \sin\left(\frac{\pi}{2} - \frac{\pi x}{12}\right)$$

$$iii) 6 - x = 24 \cos\left(\frac{\pi x}{12}\right)$$

$$\frac{6-x}{6} = 4 \cos\left(\frac{\pi x}{12}\right)$$

$$1 - \frac{x}{6} = 4 \cos\left(\frac{\pi x}{12}\right)$$

Draw $y = 1 - \frac{x}{6}$ on the same graph and read off x -values of pts of intersection.

B) 3 solⁿs

$$x \doteq -4.3$$

b) i) $\triangle BCD$ is isosceles

$\therefore \angle BDC = a$ (equal base $\angle$'s)

$\angle DCR = a + a$ (ext. $\angle$ of $\triangle BCD$)
 $= 2a$

ii) $\angle DCR = \angle DAB$ (ext. $\angle$ of cyclic quad = opp int. $\angle$)

$$\therefore \angle BAD = 2a$$

but OA bisects $\angle BAD$

$$\therefore \angle OAD = 2a \div 2 = a$$

$$iii) \angle TAD = 90 - a \text{ (OA } \perp \text{ AT)}$$

$$\angle ABD = 90 - a \text{ (}\angle \text{ in alt. seg.)}$$

$$\angle ABC = 90 - a + a \text{ (adj. } \angle \text{'s)} = 90^\circ$$